

A Participant-Driven Spatial Narrative for Cultural Exhibitions

Bok-Yin Lo

Keio University Graduate School of
Media Design
Yokohama, Kanagawa, Japan
bylo@kmd.keio.ac.jp

Kanghong Zhong

Keio University Graduate School of
Media Design
Yokohama, Kanagawa, Japan
kz6746@kmd.keio.ac.jp

Takatoshi Yoshida

Keio University Graduate School of
Media Design
Yokohama, Kanagawa, Japan
yoshida@kmd.keio.ac.jp

Dunya Chen

Keio University Graduate School of
Media Design
Yokohama, Kanagawa, Japan
dchen@kmd.keio.ac.jp

Kouta Minamizawa

Keio University Graduate School of
Media Design
Yokohama, Kanagawa, Japan
kouta@kmd.keio.ac.jp

Abstract

Traditional exhibitions often default to passive viewing, which weakens visitors' agency and attachment to cultural content. We present a projection-based, participant-driven spatial narrative for cultural exhibitions, in which each visitor choice immediately reshapes an immersive room, branching the story toward distinct outcomes. Each decision changes culturally grounded story elements and audiovisual motifs in the space. By making agency salient and consequences immediate, the design aims to increase curiosity and engagement with the theme. We report a formative in-situ study assessing agency and narrative transportation with established instruments.

CCS Concepts

• **Applied computing** → *Media arts; Interactive learning environments*; • **Human-centered computing** → **Mixed / augmented reality**.

Keywords

interactive projection, spatial narrative

ACM Reference Format:

Bok-Yin Lo, Kanghong Zhong, Takatoshi Yoshida, Dunya Chen, and Kouta Minamizawa. 2025. A Participant-Driven Spatial Narrative for Cultural Exhibitions. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771511>

1 INTRODUCTION

Cultural exhibitions aim to foster understanding of heritage and cultural knowledge, yet traditional formats often default to passive viewing, weakening visitors' agency and emotional attachment to content. Museum experience is constructed through the interaction of visitors' personal backgrounds, sociocultural contexts, and the

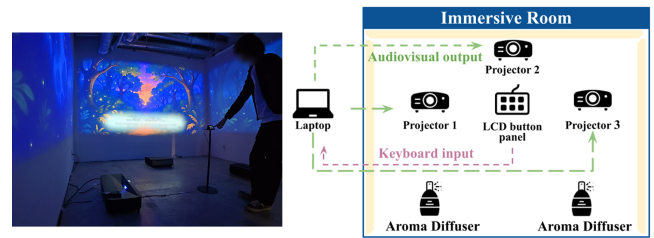


Figure 1: (a) On-site setup of the interactive 270° projection cultural exhibition pilot (b) Setup design of the system

museum's physical setting [Falk and Dierking 2016]; without opportunities for active participation that enable narrative transportation, deeper engagement and meaning-making are harder to achieve. Although projection mapping is widely used to boost engagement, many deployments remain linear and spectacle-oriented, with limited interactivity and predefined sequences loosely tied to heritage interpretation [Li and Ito 2021]. We shift emphasis from spectacle to storycraft, designing interactions in which people, space, content, and technology co-produce meaning [Wong 2015]. Grounded in spatial computing which treats computation as part of lived space and interaction [Greenwold 2003], our approach moves cultural exhibitions beyond passive spectacle.

We present a participant-driven spatial narrative, making visitors co-authors: each choice immediately reshapes the immersive room and branches the story toward distinct outcomes. Decisions map to culturally grounded story elements and audiovisual motifs on nearby surfaces. Unlike linear projection installations or button-triggered kiosks with fixed flow, the design aims to increase curiosity and engagement by making agency salient and consequences immediate. We report a formative in-situ study in a 270° projection-mapped room with spatial audio, examining links between agency and narrative transportation with established instruments.

We operationalize this in a 3 × 3 m immersive room: three floor-mounted laser projectors create a 270° projection surface and spatial audio. Signals route to a laptop outside, where Unity handles audio, visuals, and interaction. Input uses a panel with six LCD buttons, with icons mirroring on-screen options to ease learning. Two diffusers provide a light ambient scent.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771511>

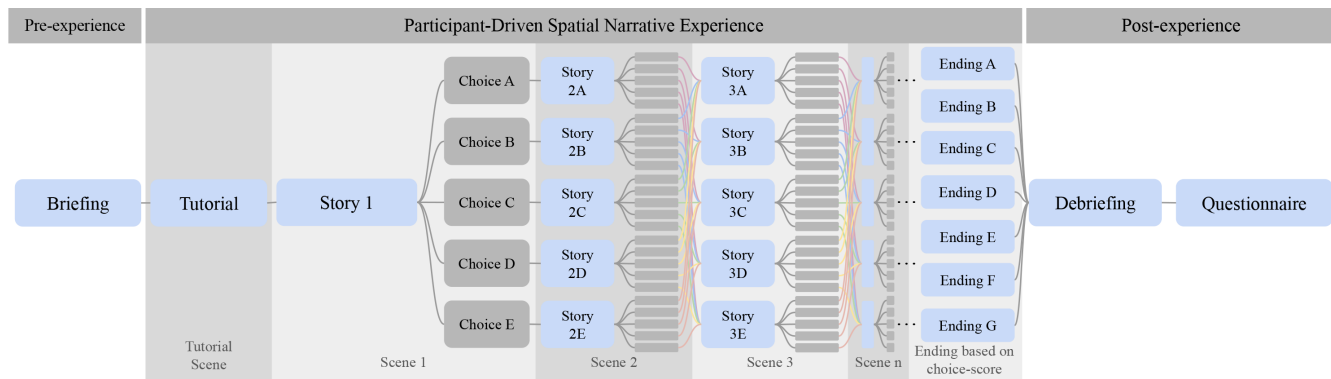


Figure 2: Configuration and processing flow of our projection system

Interaction follows a brief perceive → choose → feedback → progress loop. Each round offers five options; pressing a button immediately updates motion and sound on the nearest wall and speaker. In parallel, a scoring mechanism adds weights across narrative dimensions and determines branched endings. A debrief reveals cultural information tied to the ending and lets visitors browse other outcomes, encouraging comparison and discussion.

For visuals, we use an offline-generation, low-latency playback pipeline: we seed base images with generative AI and convert them into short, looping animations. This pipeline lets choices swap to corresponding visual motifs without breaking continuity, balancing diversity with real-time responsiveness under a limited production budget.

2 USER STUDY

We deployed the system in an art-hotel exhibition space with a Greek-mythology theme, introducing deities and their cultural contexts. After completing the interactive experience, N=27 visitors filled a 7-point Likert questionnaire and received a short debrief. We used the Transportation Scale–Short Form (TS-SF), a validated measure with strong reliability and sensitivity to story-structure manipulations, to assess narrative transportation [Appel et al. 2015]. Overall, responses indicated high ease of use and attractiveness. Narrative items such as “I was mentally involved in the story” and “I wanted to know how the story ended” received consistently high scores, suggesting strong transportation. On-site observation also showed visitors frequently sharing and comparing their branched endings and paths, prompting natural discussion—supportive of narrative autonomy and extended cultural understanding.

3 CONCLUSION AND FUTURE WORK

This work designs and deploys a participant-driven spatial narrative that turns passive viewing into a co-created cultural experience. In a 270° immersive room, each visitor’s choice immediately updates visuals and audio and advances the story. A formative in-situ pilot indicated the experience was easy to use and engaging, prompting discussion and curiosity about the exhibition’s background. We will further test effectiveness by comparing an interactive narrative with a linear baseline on the same immersive display and examining differential effects on engagement and audience response; findings

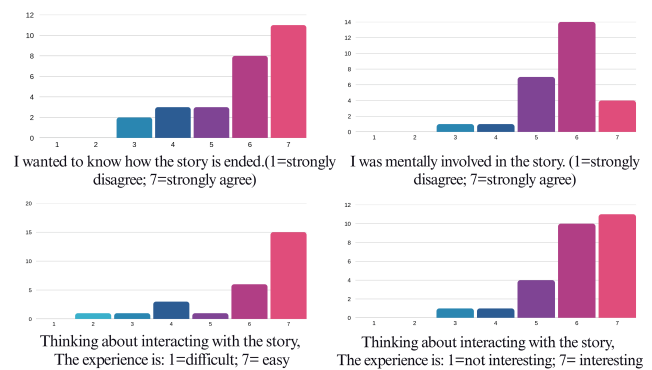


Figure 3: In-situ survey highlights (N=27; 7-point Likert). Mean ratings for Easy to Use, Attractive, Mentally Involved, and Wanted to Know the Ending were high, indicating strong narrative transportation.

will inform a reusable design/authoring framework that curators can adapt and localize. We will also study branch-curation strategies to manage authoring complexity at scale.

Acknowledgments

This project is supported by The Yoshiko Shinohara Memorial Foundation, Japan.

References

- Markus Appel, Timo Gnambs, Tobias Richter, and Melanie C. Green. 2015. The Transportation Scale–Short Form (TS-SF). *Media Psychology* 18, 2 (2015), 243–266. doi:10.1080/15213269.2014.987400
- John H. Falk and Lynn D. Dierking. 2016. *The Museum Experience Revisited*. Routledge, New York, NY. doi:10.4324/9781315417851
- Simon Greenwold. 2003. *Spatial Computing*. Master’s thesis. Massachusetts Institute of Technology, Program in Media Arts and Sciences, Cambridge, MA. <https://acg.media.mit.edu/people/simong/thesis/SpatialComputing.pdf>
- Hengyi Li and Hiromu Ito. 2021. Comparative analysis of information tendency and application features for projection mapping technologies at cultural heritage sites. *Heritage Science* 9, 1 (October 2021), 126. doi:10.1186/s40494-021-00579-0
- Amelia Wong. 2015. The Whole Story, and Then Some: ‘Digital Storytelling’ in Evolving Museum Practice. In *Museums and the Web 2015*. Chicago, IL. <https://mw2015.museumsandtheweb.com/paper/the-whole-story-and-then-some-digital-storytelling-in-evolving-museum-practice/>